

Data Cleaning and Preparation using Python

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✎ Import the Pandas library for data analysis and manipulation

```
import pandas as pd
```

✎ Load the dataset into a Pandas DataFrame

```
df = pd.read_csv('/content/train.csv', encoding='latin1')
```

✎ Display the first 5 rows of the dataset

```
df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region
0	1	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South
1	2	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South
2	3	CA-2017-138688	12/06/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036.0	West
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311.0	South
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311.0	South

✎ Display dataset information including columns and data types

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Row ID              9800 non-null  int64
1   Order ID            9800 non-null  object
2   Order Date          9800 non-null  object
3   Ship Date           9800 non-null  object
4   Ship Mode           9800 non-null  object
5   Customer ID         9800 non-null  object
6   Customer Name       9800 non-null  object
7   Segment             9800 non-null  object
8   Country             9800 non-null  object
9   City                9800 non-null  object
10  State               9800 non-null  object
11  Postal Code         9789 non-null  float64
12  Region             9800 non-null  object
13  Product ID          9800 non-null  object
14  Category            9800 non-null  object
15  Sub-Category        9800 non-null  object
16  Product Name        9800 non-null  object
```

```
17 Sales          9800 non-null float64
dtypes: float64(2), int64(1), object(15)
memory usage: 1.3+ MB
```

✓ Check for missing values in each column

```
df.isnull().sum()
```

	0
Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country	0
City	0
State	0
Postal Code	11
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0

dtype: int64

✓ Remove rows containing missing values

```
df = df.dropna()
```

✓ Check duplicate values

```
df.duplicated().sum()
```

```
np.int64(0)
```

✓ Remove duplicate rows from the dataset

```
df = df.drop_duplicates()
```

✓ Convert Order Date column into datetime format

```
df['Order Date'] = pd.to_datetime(df['Order Date'], dayfirst=True)
```

✓ Convert Ship Date column into datetime format

```
df['Ship Date'] = pd.to_datetime(df['Ship Date'], dayfirst=True)
```

✓ Verify updated data types and cleaned dataset information

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 9789 entries, 0 to 9799
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype  
---  --
0   Row ID                 9789 non-null  int64   
1   Order ID               9789 non-null  object  
2   Order Date             9789 non-null  datetime64[ns]
3   Ship Date              9789 non-null  datetime64[ns]
4   Ship Mode              9789 non-null  object  
5   Customer ID            9789 non-null  object  
6   Customer Name          9789 non-null  object  
7   Segment                9789 non-null  object  
8   Country                9789 non-null  object  
9   City                   9789 non-null  object  
10  State                  9789 non-null  object  
11  Postal Code            9789 non-null  float64  
12  Region                 9789 non-null  object  
13  Product ID             9789 non-null  object  
14  Category               9789 non-null  object  
15  Sub-Category           9789 non-null  object  
16  Product Name           9789 non-null  object  
17  Sales                  9789 non-null  float64  
dtypes: datetime64[ns](2), float64(2), int64(1), object(13)
memory usage: 1.4+ MB
```

✓ Export cleaned dataset to a new CSV file

```
df.to_csv('Cleaned_train.csv', index=False)
```